Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

Ringkasan. This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[fleqn,10pt]ncc

1]First Author 2]Second Author [1]Department, University, City, Country of first author [2]Department, University, City, Country of first author [1]First Authorf.author@email.com

Keyword1, Keyword2, Keyword3, Keyword 4, Maximum 5 keywords

Ringkasan. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. Dummy abstract text. No more than 200 characters.

Article Title Example

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1. Introduction

Your introduction goes here! Some examples of commonly used commands and features are listed below, to help you get started.

If you have a question, please use the help menu in the top right of the screen to get in touch. When your article or pre-print is complete, use the "Submit to NCC" button in the topbar to send your files to NCC.

1.1. About NCC

Novel Computation and Control is an award-winning open access publisher covering the biological and medical sciences. NCC provides authors with three publication venues: *NCC* and *NCC Computer Science* (peer-reviewed academic journals) and *NCC PrePrints* (a 'pre-print server').

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2. Some LaTeX Examples

Use section and subsection commands to organize your document. LaTeX handles all the formatting and numbering automatically. Use ref and label commands for cross-references.

2.1. Figures and Tables

Use the table and tabular commands for basic tables — see Table 1, for example. You can upload a figure (JPEG, PNG or PDF) using the project menu. To include it in your document, use the includegraphics command as in the code for Figure 1 below.

Gambar 1. An example image.

Item	Quantity
Widgets	42
Gadgets	13

Tabel 1. An example table.

2.2. Citations

LaTeX formats citations and references automatically using the bibliography records in your .bib file, which you can edit via the project menu. Use the cite command for an inline citation, like [?], and the cite command for a citation in parentheses [?, ?, ?, ?].

2.3. Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

2.4. Lists

You can make lists with automatic numbering ...

[noitemsep]Like this, and like this.

... or bullet points ...

[noitemsep]Like this, and like this.

... or with words and descriptions ...

Word Definition

Concept Explanation

Idea Text

We hope you find writeLaTeX useful for your NCC submission, and please let us know if you have any feedback. Further examples with dummy text are included in the following pages.

3. Methods

[4]

$$\cos^3\theta = \frac{1}{4}\cos\theta + \frac{3}{4}\cos 3\theta \tag{1}$$

[5]

3.1. Subsection

[6]

Paragraph [7]

Paragraph [8]

Gambar 2. In-text Picture

3.2. Subsection

[9]
Reference to Figure 2.

4. Results and Discussion

[10]

4.1. Subsection

[11]

4.1.1. Subsubsection

[12]

4.1.2. Subsubsection

[14]

4.2. Subsection

[15-20]

Acknowledgments

The authors of this paper are partially supported by the US national grant No. 123456.